

From a Correct Simulator to a Ladder Player

ABRA on Pokémon Champions VGC 2026 Reg M-C, open team sheets

Version 1.0.0 · Last updated 2026-09-24 Line: abra/regmc — [CHANGELOG-REGMC.md](#) .

1.0.0 — THE SIMULATOR IS CERTIFIED ON REG M-C, SO THE SEARCH CAN BEGIN. [node](#)

[engine/quarantine.js --regulation regmc](#) reads OPEN on every gating clause on release [eaa5becc54eb](#) . That was the foundation. This paper is about what is built on it: a player that solves each turn of a Reg M-C doubles battle as a simultaneous-move game, inside our own copy of the game, fast enough to play on the Showdown ladder under the tournament clock. The finish line is a settled rating high on the [gen9championsvgc2026regmcbo3](#) ladder, played by the account [medicham32](#) , which Showdown staff have approved. **No solver result in this paper is a ladder result, and no strength figure is published.** Every solver measurement so far is either OFFLINE (fitted and scored on recorded human games, with no simulator in the path) or PRE-GATE (played inside a simulator that had not yet passed its gate). The offline figures are given with their receipts. The PRE-GATE figures are withheld until they are re-run on the certified release (§3.8).

WHAT THIS EDITION REMOVED. This paper was the Reg M-B white paper until 1.0.0. Reg M-B is retired (Will, 2026-09-24). Its published record is the 7.0.0 edition, which stays exactly as it was written and is read with `git show bfbf9cf9:docs/ABRA-whitepaper.md` . Every model figure that edition carried outside the simulator had already been withdrawn at 7.0.0, and the models themselves are now being rebuilt from scratch. So those figures are deleted here, not captioned. The Reg M-B simulator figures appear once, in §5, as the OLD SERIES of the basis-change comparison, labelled as Reg M-B.

Abstract

Pokémon VGC doubles is a two-player game with simultaneous moves, a large joint action space, heavy chance and a small amount of hidden information. Under open team sheets the moves, items and abilities of all six opposing Pokémon are public. What stays hidden is which four of the six were brought, and the stat spreads, which decide speed order and damage. ABRA's plan is to search this game at decision time rather than to learn a policy that plays without search. At each turn it builds a matrix game between a pruned set of its own joint actions and the opponent's, averaged over the hidden worlds it still believes possible. It fills each cell by playing the position forward inside MEDICHAM, its own simulator, with shared dice across cells, and it solves the matrix for an equilibrium mix (SLOWKING). Human-behaviour models narrow the candidates (MAG, DODUO) and fill in the hidden information (XATU). A value network (PORYGON2), trained by a self-play loop (MEW, MACHAMP), later replaces the rollout at the leaf. A capped exploit dial (HYPNO) moves off the equilibrium only as far as a computed bound allows, and an exploiter (WOBBUFFET) measures how beatable each version is. Team preview is its own matrix game (CHOMP). A live client (ROTOM) plays it on the ladder under Showdown's clock. The whole plan rests on one fact, which this release establishes for Reg M-C: MEDICHAM plays the same game as the official simulator, clause by clause, on every instrument the project owns.

1. The thesis: the engine is the foundation, the search is the point

A search is worth exactly what its model of the game is worth. A simulator that plays a slightly different game produces confident answers about a game nobody is playing, and nothing in the search itself can notice. That is why the project spent its first two regulations on MEDICHAM and published nothing downstream of it until

a gate certified it (§4). With the gate open on Reg M-C, the order of work reverses: MEDICHAM is frozen as the foundation, and the effort goes to the search.

Why search, not a policy that plays without it. Three reasons, from the research record

(docs/_reports/2026-09-23-solver-research-turn-search.md §2.4):

- 1. Reg M-C went live on 2026-09-09. Its human corpus is orders of magnitude smaller than the corpora that trained the strongest no-search agents, which also play singles, not doubles.
- 2. The closest prior work, PokaiTrainer (Yu, arXiv 2608.29197), played this exact problem on Reg M-B with open sheets. It reports that search beat every raw policy head it trained, and that every search agent beat its own policy head [REPORTED, not reproduced].
- 3. A laptop can afford decision-time CPU search. It cannot afford the self-play volume that the no-search equilibrium methods need.

The metric. The headline is a settled ladder rating with its standard deviation and the number of series behind it, read over the last N series with N at least one hundred. It is never the peak. A peak is the maximum of a noisy walk and sits well above the true rating (docs/_reports/2026-09-23-solver-research-humans-and-ladder.md §3.1). Offline, each model must beat a named baseline in a pre-registered test before it may be promoted (§3). A strength claim between two versions of the agent is a sequential probability ratio test (SPRT) at equal wall-clock, on paired seeds and the same team pairs.

2. The search plan

Plan version 0.2.0 is solver/PLAN.md . The running log is solver/LOG.md . This section states the design. §3 states what exists and what has been measured.

2.1 The game as the searcher sees it

Property	What it forces on the design
Both players commit at the same time	The root is a matrix game. Its answer is an equilibrium MIX, not an argmax. A deterministic best response is exploitable by construction.
Each side chooses a PAIR of actions	The joint action count is roughly the per-slot count squared. Pruning is mandatory, and it must be measured by JOINT coverage, because per-slot coverage overstates it.
Damage rolls, accuracy, critical hits, secondary effects, speed ties	Cell values are noisy. The noise between cells must be paired (common random numbers), or the matrix measures dice rather than actions. A speed tie is a branch to plan both ways, not one die to sample.
Hidden information is small under open sheets	What stays hidden is which four of six were brought — $C(6,4)$ worlds at preview, $C(4,2)$ once both leads are seen — plus the spreads. That is small enough to enumerate worlds rather than learn an abstraction.
A chess clock with a per-turn cap	The search must be anytime: stoppable at any moment with a solvable matrix. §2.10 gives the clock.

2.2 The pipeline

Team preview runs once per game, before the first turn:

both open sheets

→ CHOMP

every bring-and-lead option per side: $C(6,4) \times C(4,2)$ of them

→ PORYGON2

score each cell of the preview matrix (the rollout leaf until PORYGON2 exists)

→ SLOWKING

solve the preview game → a mixed strategy → sample the bring and the lead

Then every turn:

```
request + |inactive| clock line
```

- XATU worlds: the back-two posterior $\times$ spread candidates, reweighted by what has been seen
- MEDICHAM legalActions per side, per world (the engine's own menu, never a hand-built one)
- MAG/DODUO prune to k1 own and k2 opponent joint actions, with switch and mega slots reserved
- MILTANK fill cells: clone the state, step one turn with shared dice, score the leaf
(truncated rollout now, PORYGON2 later); spend playouts by successive halving (planned)
- SLOWKING regret matching on the belief-averaged matrix $\rightarrow \sigma_{me}, \sigma_{opp}, \text{value } v^*$
- HYPNO move toward GARY's model of human habit, never further than the ϵ cap allows
- sample from σ_{play} (never argmax) $\rightarrow \log$ every counter

The per-turn pipeline follows `docs/_reports/2026-09-23-solver-research-turn-search.md` §3.1. Each name is a model with one job, one input and one output (§3.1 registry).

2.3 SLOWKING — the per-turn equilibrium

For a pruned own set A , a pruned opponent set B and a belief β over worlds w , the cell value is

$$M[a][b] = \sum_w \beta(w) \cdot E[\text{leaf}(\text{step}(S_w, a, b))],$$

where S_w is the current state completed with world w , `step` is one MEDICHAM turn under shared dice, and the expectation is over playouts. SLOWKING solves max over σ_{me} of min over σ_{opp} of $\sigma_{me}^T M \sigma_{opp}$ by regret matching plus (RM+, Tammelin 2014), which has an average-strategy exploitability bound of order $\Delta(\sqrt{m} + \sqrt{n})/\sqrt{T}$ for an $m \times n$ matrix with payoff range Δ after T iterations. It also carries an exact linear-programming solve, used to check RM+ on small matrices. Solving costs microseconds against the cost of filling cells, so the matrix is always solved; only the cells are budgeted.

Two later stages are planned and not built. **Per-world opponent tables:** the opponent knows its own world, so a single public opponent strategy under-models it (the strategy-fusion error of Long et al. 2010, which runs against the opponent here, not against us). The fix is one opponent regret table per world, solved by alternating-update linear CFR, as PokaiTrainer does. **Depth two:** simultaneous-move MCTS below the root with regret-matching selection, which converges (Lisý et al. 2013). It is built only if a measurement shows that the leaf, not the width, limits play.

2.4 MILTANK — filling the matrix under a clock

MILTANK is the harness around SLOWKING. It asks DODUO for the candidate joints, is to ask XATU for the worlds (v1 draws them uniformly, §3.5), forks the engine's own state through the MEDICHAM API (never a lossy re-seed from a feature board, which is what broke the Reg M-B harness), and fills cells:

- **Common random numbers.** The i -th playout of every cell draws the same event-addressed dice, so a difference between two cells measures the actions, not the luck.
- **Successive halving over MY rows** (planned; not in v1) spends the playout budget where it can still change the answer. It never halves over the opponent's columns, because the equilibrium needs those columns filled.
- **Reserved slots.** A share of each shortlist is kept for switch-containing and mega-spending joints, because a cheap screen always under-rates them.
- **Anytime.** Cells are filled in priority order. At the deadline the partial matrix is solved, an unfilled cell takes the prior's value, and the count of unfilled cells is logged.
- **Fallback that is counted.** If the search throws or runs out of time, the client plays MAG's top legal action and increments a counter. A run with zero fallbacks has to be shown, not assumed.

2.5 MAG and DODUO — the human policy prior that prunes the tree

MAG scores each slot's options from the public state and both sheets. DODUO scores the PAIR as one joint action, so that a focus-fire, a redirect-then-attack or a double switch is valued as the coordinated choice it is. Both are fitted on human Reg M-C open-sheet games only — no simulator is in the path — and are split by player, so the test players were never seen in training. The pruner's gate is joint coverage on held-out human turns, with its threshold set before the run.

2.6 XATU — the belief over what is hidden

XATU keeps a posterior over the opponent's back two and over each body's spread. The back-two prior is the store's bring frequencies for that sheet, updated deterministically on every reveal. The spread posterior is reweighted each turn by the likelihood MEDICHAM's own transition assigns to what was seen: turn order is a near-hard constraint away from speed ties, and an observed damage percentage is checked against the roll range from the engine's exported `dmgRange`. The same function prices damage on both sides of that comparison, because a fact computed twice diverges. The belief prefers what it has OBSERVED to what the sheet DECLARED: a removed or consumed item is tracked per body.

2.7 PORYGON2, MEW and MACHAMP — the value network and the loop that trains it

A truncated random rollout is the first leaf because it needs nothing trained. PORYGON2 replaces it: a network $v_\theta(\text{public state, world})$ trained on search targets from self-play, mixed with game outcomes. MEW is the self-play factory: it plays games on a frozen MEDICHAM release and stamps every record with the release id and the weights digest, so that data from a moved engine is dropped or down-weighted, never mixed silently. MACHAMP is the training loop: expert iteration, with a league that keeps a human-behaviour anchor so self-play does not drift into a game humans do not play. Training is in Python, inference in hand-written Node, and a JavaScript-against-Python parity test must go red on a perturbation before it is trusted.

PORYGON2 is promoted only up a ladder of tests, each against a named baseline (`solver/PLAN.md` Appendix B):

Rung	Question	Must beat
Vo	Better than material?	a logistic model on the difference in bodies alive and in HP
V1	Does it know the engine's game?	the same baseline, against MEDICHAM rollout values
V2	Does it help the search?	the same search with the material leaf, by SPRT at equal wall-clock
G1	Is the next generation better?	the previous checkpoint, by SPRT on the same release
W	Did it run?	counters for leaf evaluations, prunes and fallbacks — a zero fails

Turns one to three are always reported separately: the closest prior work found its value network stayed optimistic early in the game [REPORTED].

2.8 HYPNO and GARY — a capped exploit dial

An equilibrium mix cannot lose much to anyone, and it also does not take points from predictable opponents. HYPNO trades a bounded amount of the first property for some of the second. GARY models what humans actually click in a given situation bucket. HYPNO then plays an ϵ -safe response to GARY's model: the worst case of the played mix must stay within ϵ of the equilibrium value v^* on every decision, the Restricted Nash Response construction of Johanson, Zinkevich and Bowling (2007). The trust placed in a bucket grows with its data, $p(\text{bucket}) = p_{\text{max}} \cdot n / (n + K)$. The starting cap is half a win-probability point per decision (Will, 2026-09-24). A GARY bucket that does not beat the equilibrium prediction on held-out games gets $p = 0$.

2.9 WOBBUFFET and DUSK — measuring the holes, closing the endgames

WOBBUFFET is trained against a frozen version of the agent to find what beats it. Its win rate must not rise from one generation to the next; if it does, the new version is more exploitable even if it beats the old one head to head. DUSK computes exact values for small late-game positions, so the search does not estimate what can be solved.

2.10 The clock

The Reg M-C Bo3 format is [Gen 9 Champions] VGC 2026 Reg M-C (Bo3) , whose ruleset is Flat Rules , VGC Timer , Force Open Team Sheets and Best of = 3 (pokemon-showdown-mc/config/formats.ts:295-299 , checkout f10d679). The timer rule reads, verbatim (pokemon-showdown-mc/data/rulesets.ts:778-785 ; the Champions mod's own timer line is commented out):

```
vgctimer: {
  desc: "VGC's timer: 90 second Team Preview, 7 minutes Your Time, 1 minute per turn",
  ruleset: [
    'Timer Starting = 420', 'Timer Grace = 90',
    'Timer Add Per Turn = 0', 'Timer Max Per Turn = 55', 'Timer Max First Turn = 90',
    'Timeout Auto Choose', 'DC Timer Bank',
  ],
}
```

So each player has a bank of Timer Starting = 420 seconds with no increment, and no single request may take more than Timer Max Per Turn = 55 seconds. The unit of spend is a REQUEST, and a replacement after a knock-out draws on the same bank. The client parses the |inactive| line on every request and budgets $budget = \min(X - margin, (bank - reserve) / E[remaining requests])$, where X is the time left this turn and E[remaining requests] is read from the store as a function of turn number and bodies left — never typed. The reserve is asymmetric on purpose: an expired turn concedes one server-chosen move, but an empty bank forfeits the game. The search workers are started at connect time and warmed during team preview, because a fresh Node process runs MEDICHAM several times slower until the JIT tiers up (warm_state_effect in the speed record the research report cites).

2.11 ROTOM — the live client

ROTOM replaces engine/mag_bot.js , which only accepted challenges and never searched the ladder. It plays gen9championsvgc2026regmcbob3 only — the bob1 ladder is closed-sheet and out of scope. It reads its password from a git-ignored file or the environment, never the command line. It plays one series at a time, rotates a small set of real top ladder teams per series (Will, 2026-09-24), and stops on a kill-switch file. It stays connected through a Bo3 series, because the rating updates once per series and a rename forfeits it, and it carries game-one observations into games two and three. It rejoins every open room after a disconnect and re-reads the last request before choosing. A machine-wide lock refuses a ladder search while one of Will's own accounts could be laddering the same format. It logs one record per series: arm, ε, both players' ratings before and after, score and expected score, the number of HYPNO deviations, clock used, the engine release and the weights digests.

2.12 How a version is judged

Layer	What	Rule
In-process arena	Both agents play inside MEDICHAM through the API	A frozen release, a census pin and the frozen Reg M-C pool; --games and every flag recorded

Layer	What	Rule
Local-server arena	Both agents play through ROTOM on a local Reg M-C server, timer on	Proves the client and the clock, not strength: zero timeouts, zero bank forfeits
Head-to-head	SPRT at equal wall-clock, paired seeds, the same team pairs	Promotion only on a pass
Exploitability	Root exploitability logged on every decision; WOBBUFFET against the frozen agent	The exploiter's win rate must not rise
Ladder	One account; the arm is drawn per series by a seeded coin committed in advance; the metric is the mean per-series residual $S - E$	Headline: mean rating over the last N series $\pm$ SD, N at least one hundred. Never the peak

Rating noise on the ladder is large: the research report derives the stationary spread of a rating under the ladder's update rule and the number of series needed to resolve a given gain (docs/_reports/2026-09-23-solver-research-humans-and-ladder.md §3.1 to §3.3). The practical conclusion: a large gain can be read on the ladder in weeks, and a small one must be settled offline.

2.13 Milestones

M	What lands	Exit test	Status
M0	Reg M-C MEDICHAM gate open; the solver API merged	the gate reads OPEN; the API's legality probe agrees with the authority	Gate OPEN at 1.0.0. The API is merged. The mid-turn choice callback (a faint replacement or pivot switch-in asked, not pre-decided) is not built
M1	DODUO split from MAG; XATU bring posterior; GARY buckets; MAG intersected with legalActions	DODUO beats the no-pair model; XATU beats store bring frequencies	MAG, DODUO and XATU v1 built offline; GARY not built
M2	SLOWKING + MILTANK: one-ply matrix, rollout leaf, reserved slots	solver unit tests; joint coverage over the pre-set bar; SPRT against greedy one-turn and against MAG with no search	built; strength not measured on the certified release
M3	Clock and ROTOM on a local server; the offline arena	timer-on series with zero timeouts and zero bank forfeits; a disconnect drill	arena built; ROTOM not built
M4	CHOMP v0; ALAKAZAM v1; the first ladder burn-in at $\varepsilon = 0$	preview head-to-head against the human-modal bring; the per-series residual within one SE of zero	not started
M5	PORYGON2 on MEW self-play; the MACHAMP loop	$V_0 \rightarrow V_1 \rightarrow V_2$; G1 passes twice	not started
M6	Per-world opponent tables; WOBBUFFET; DUSK	a toy Bayesian game solved exactly; SPRT at equal wall-clock; exploiter win rate not rising	not started
M7	HYPNO + GARY on the ladder	ε Pareto sweep offline; per-series randomised ladder A/B	not started
M8	CHOMP v1, DITTO (team builder), the GURU replicator test, KADABRA (coach)	CHOMP v1 beats v0; DITTO survives the pilot-confound check	not started

3. What exists, and what has been measured

3.1 The model registry

Every model except MEDICHAM is new code under solver/ , written from scratch for Reg M-C and open sheets. The old Pokémon names are kept for the roles solver/PLAN.md §2 gives them. SEARCH becomes the SOLVER division, which owns solver/ and every model below except MEDICHAM.

Model	Role	Status on 2026-09-24
MEDICHAM	The simulator and its solver API	Certified on Reg M-C (1.0.0). The only old model kept
MAG	Per-slot action scorer: the human policy prior	v1 built, offline
DODUO	Joint coordinator: scores the pair as one action	v1 built, offline
XATU	Belief over the back two and the spreads	v1 built, offline
SLOWKING	Per-turn simultaneous-move solver	v1 built; unit-tested
MILTANK	Search harness: candidates, shared-dice playouts, clock (halving and XATU worlds planned)	v1 built; strength PRE-GATE, withheld
PORYGON2	Value network	to build
MEW	Self-play factory on a frozen release	to build
MACHAMP	Training loop and league	to build
GARY	Human habit per situation bucket	to build
HYPNO	Capped exploit dial	to build
WOBBUFFET	Exploitability best-responder	to build
DUSK	Endgame tables	to build
CHOMP	Team-preview solver, rebuilt inside ABRA under <code>solver/chomp/</code>	to build. The old CHOMP repository runs on stale mainline data and is reference only
JOLTEON	Optional fast pre-screen for CHOMP's preview matrix	built only if PORYGON2 is too slow to score every cell
ROTOM	The live ladder client	to build
ALAKAZAM	The assembled agent	to build
GURU	Descriptive meta: usage, sets, archetypes, bring and lead, Bo3 adaptation	vo exists, descriptive
DITTO	Team builder	to build
KADABRA	Coach: explains ALAKAZAM's logged decisions	to build

3.2 The human dataset and the descriptive meta — OFFLINE

The human-play dataset parses the Reg M-C Bo3 store into turns and side-turn decisions with the full joint action where it is visible, excluding named and behavioural bots, custom-rule rooms, the Illusion exclusion and games played before the Eject Button authority fix (`docs/_reports/2026-09-23-human-dataset.md`). None of Will's accounts is in it, and `medicham32` is on the exclusion list of both the dataset and the meta. The descriptive meta (GURU vo) finds that games cluster strongly by player, so every interval it reports is clustered by player; its archetypes did not meet their pre-set stability bar and are treated as rough (`docs/_reports/2026-09-23-regmc-meta.md`). Neither reads MEDICHAM, so neither is quarantined.

3.3 MAG v1 and DODUO v1 — OFFLINE (abra/regmc 0.113.0)

Fitted and scored on human games only, split by player. On the held-out test players (25,477 exact joint actions), DODUO's joint log-loss is 2.730, 95% CI 2.699–2.761, against 2.921 for the vo prior it replaces (CHANGELOG-REGMC 0.113.0, read from `solver/mag/model/mag-doduo-v1.metrics.json`). The Node forward pass matches Python to 2.1e-14, and `solver/tests/test-mag-doduo.js` passes 3,826 checks after going red on deliberate breaks first. The weakest slices remain switch turns and turn one (`docs/_reports/2026-09-24-mag-doduo-v1.md`). This measures how well the models PREDICT humans. It says nothing yet about how well they PRUNE for a search; that is the joint-coverage gate of M1.

3.4 XATU v1 — OFFLINE (abra/regmc 0.114.0)

On 10,942 held-out sides, XATU's turn-one log-loss on the true back pair is 1.348 against 1.792 for a uniform belief, and the true bring is never ruled out — zero of 10,942 (CHANGELOG-REGMC 0.114.0, docs/_reports/2026-09-24-xatu-v1.md). Spread narrowing is weak: most candidate values survive a game. That is the part of XATU that reads MEDICHAM's damage function, and it is the part to improve.

3.5 SLOWKING v1 and the MILTANK v1 skeleton — built (abra/regmc 0.115.0)

solver/tests/test-slowking.js passes 1,559 checks: rock-paper-scissors solves to uniform, known small games solve to their linear-programming answer, RM+ stays under its proven bound, and dominant actions are found in constructed fixtures. MILTANK v1 draws the opponent's world uniformly and has no successive halving; wiring XATU's posterior in and halving over the cells are the next steps (solver/PLAN.md §2). test-miltank.js passes 3,414 and test-arena.js 15, each shown red on deliberate breaks first (CHANGELOG-REGMC 0.115.0).

3.6 Playouts: a worker pool (abra/regmc 0.116.0)

A worker-process pool fills MILTANK's cells bit-identically to the serial path at a pass cap. Before cut-short passes were started at golden-ratio offsets, the pool left 34.8% of cells empty at a one-second budget (CHANGELOG-REGMC 0.116.0). test-playout-speed.js passes 1,184 checks. About four-fifths of a playout is the engine's own turn, so the engine's speed is the search's speed (docs/_reports/2026-09-24-playout-speed.md).

3.7 Lean playouts (abra/regmc 0.116.1)

A lean battle mode runs the same turn with no protocol text, no counters and tag answers from a table. It is proven board-identical to a full battle on every game of the Reg M-C --games 1200 lattice (955 games, 22,283 turns) and on 300 human-sheet games (2,518 turns), and the test goes red on a deliberate break. Per MILTANK playout it is 1.25x to 1.33x faster in paired blocks of main-thread CPU, with the cell values hashed equal (CHANGELOG-REGMC 0.116.1, docs/_reports/2026-09-24-lean-mode.md).

3.8 The arena shakedowns — PRE-GATE, figures WITHHELD

MILTANK has played shakedown matches in the offline arena against a greedy player that takes the prior's top joint action (docs/_reports/2026-09-24-slowking-miltank-arena.md , docs/_reports/2026-09-24-playout-speed.md). **Their win rates are not printed here.** They were played inside a MEDICHAM that had not passed its Reg M-C gate, and the arena reads the live engine tree rather than a frozen release. Under this project's quarantine rule a figure downstream of an uncertified simulator is withheld, not captioned — the gate opening makes it RE-RUNNABLE, not true. The shakedowns did establish two engineering facts that do not depend on the simulator being right: the first harness was starved of playouts at a one-second budget, and cheaper playouts (§3.6, §3.7) came before any strength claim. **Owed before a strength figure is published:** the arena pinned to a frozen release at or after eaa5becc54eb , a census pin and the frozen pool, re-run at equal wall-clock, and read by SPRT.

4. The foundation: MEDICHAM certified on Reg M-C (abra/regmc 1.0.0)

4.1 What the gate requires

engine/quarantine.js --regulation regmc reads only Reg M-C artifacts (data/<name>-regmc.json) and the frozen pool data/team-pool-frozen-regmc . A missing, stale or duplicate artifact reads CANNOT-ANSWER, never a pass. The whole-game clause plays three team lattices, because --games selects WHICH teams play,

not how many of the same: a zero on one lattice is a fact about that lattice. For each lattice ℓ the gating quantity is

$$B_\ell = \text{state.games} - \text{state.games_board_never_diverged},$$

and the gate is open only when every B_ℓ is zero, no game threw, the undeclared narration-only count N_ℓ is zero on every lattice, and every other clause passes. The bar is the BOARD — commentary may differ, boards may not — and narration is a second gate of its own (Will, 2026-08-22).

4.2 The reading

Release eaa5becc54eb , census pin 123aa264f88d , pool data/team-pool-frozen-regmc , lattices --games 1200 , --games 1600 and --games 1900 , flags --steering empirical --arm middle --end-state . Account: docs/_reports/2026-09-24-regmc-gate-final.md .

Clause	Reading on eaa5becc54eb
Damage differential	0 disagreements of 6000 at the midpoint, the top, the bottom and all 14 interior roll indices, seed 20260804 (data/engine-diff-regmc.json)
Roster, items	166 of 166 boards match; the red demonstrations bite, 28 of 28 (data/roster.items-regmc.json)
Roster, abilities	210 boards match of 214 in scope (data/roster.abilities-regmc.json)
Roster, moves	510 boards match of 511 in scope (data/roster.moves-regmc.json)
Whole game, lattice 1200	955 games, 954 compared and 1 void, no board parts (data/game-differential-regmc.json)
Whole game, lattice 1600	1266 games, no board parts (data/game-differential.g1600-regmc.json)
Whole game, lattice 1900	1497 games, no board parts (data/game-differential.g1900-regmc.json)
Narration	zero undeclared on every lattice; the baseline is stamped at zero on the 955-game lattice (data/whole-game-baseline-regmc.json)
Staged mechanics	4867 games, none threw, none diverged (data/all-mechanics-fire-regmc.json)
Census	1024 rows live of 1024 probed (data/mechanics-census-regmc.json)
Open defects	no open register row names a RED instrument; 205 verdicts read (data/register-reality-regmc.json)

Of the four abilities in scope and not matched, three are announcement-only on receipts and one is Illusion, the declared exclusion. The move not matched is deferred by its owner (CHANGELOG-REGMC 1.0.0). Coverage: every move above 25 clicks is measured.

What the gate does not include, stated. Reg M-B's certification was followed by a wide held-out draw, an instrument the gate does not run, which on Reg M-B had caught what the lattices could not (§5). **No held-out draw has been run on Reg M-C.** It is owed, and it is the first measurement to make on eaa5becc54eb .

4.3 How it got here: the Reg M-C line from 0.1.0 to 1.0.0

This is the fold-in of every running-notes row on the abra/regmc line. The row-by-row record is docs/RUNNING-NOTES.md and CHANGELOG-REGMC.md .

- **A second authority (0.1.0).** A second Showdown checkout, pokemon-showdown-mc at f10d679 , so that pulling Reg M-C can never move the bytes Reg M-B's figures rest on. The delta against the pinned Reg M-B authority: species +35, moves +15, items +18, abilities +0, nothing removed — 41 new mechanics (15 abilities, 14 moves, 12 held items), not 35 species. Two moves legal in both regulations had their PP cut, and one item was unbanned: changes no added-or-removed list reveals.
- **The regulation is a flag (0.10.0 – 0.17.0).** One resolver selects the checkout, the species table, the census and the gate. The gate answers per regulation from 0.17.0.

- **The new mechanics, each tagged from its handler (0.16.0 – 0.43.0).** Each item and ability new to Reg M-C gets a tag derived from its Showdown handler, a probe, and a census row.
- **The lab and the pool (0.3.0 – 0.79.0).** The frozen Reg M-C team pool, a per-regulation census and roster, and every refused roster fixture rebuilt legal rather than scoped out.
- **Engine passes against the lattices (0.40.0 – 0.94.0).** Each fix a defect the differential or a staged mechanic found, most in both regulations. After pass ten the gate read CLOSED on five of 10 clauses with boards already at zero (0.87.0).
- **The solver API and the move menu (0.95.0 – 0.100.0).** `engine/medicham_api.js` exposes clone, an RNG handle, `legalActions`, a non-mutating `step` and a terminal test. `legalActions` agreed with the authority on 5,526 of 5,552 slots; the 26 were the engine's own move menu, and three menu fixes brought the disagreement to zero at `--games 45` (0.96.0 – 0.98.0).
- **The last reds (0.101.0 – 0.112.0).** Roster staging restored, register instruments that answer under Reg M-C, and a merge pass after which each gate failed on one clause (0.112.0).
- **The solver lands (0.113.0 – 0.116.1)** – §3.3 to §3.7.
- **Narration to zero (0.117.0 – 0.123.0).** The last commentary differences were closed one by one until the three lattices read zero undeclared.
- **The gate (0.124.0 – 1.0.0).** An item-roster red demonstration re-aimed and the census republished (0.124.0); the full re-read OPEN (0.125.0); declared as this release.

5. The basis change: from Reg M-B to Reg M-C (abra/regmc 1.0.0)

5.1 Why this is a major revision

A major version here is a change of BASIS: the question a published number answers changes, so an old and a new figure cannot be linked by a sentence of the form "X became Y". This follows the ESS *Guidelines on Revision Policy* (Eurostat, 2013, `KS-RA-13-016`), which separate a routine revision from a major one — "a change in the concepts, definitions and/or classifications used to produce the series" (Item 3.0) — and require the major one to ship with its reasons, its impact, and "a comparison between the 'new' and the 'old' series" (Item 3.4). Two things change at once here:

1. **The regulation.** Every figure now answers a question about Reg M-C, on the Reg M-C authority, over a Reg M-C pool. Reg M-B is retired (Will, 2026-09-24) and its line is CLOSED at 7.0.0.
2. **What may be published.** The gate opening lifts the quarantine on the Reg M-C line: the downstream artifacts become RE-RUNNABLE on `eea5becc54eb`, and nothing measured before that release may be quoted until it is re-run (CHANGELOG-REGMC 1.0.0).

5.2 Old series against new series

The simulator's certification is the one series that exists on both lines. The old series is Reg M-B at 7.0.0, release `0d7b1d9db6d1`, on the pinned Reg M-B authority. The new series is Reg M-C at 1.0.0, release `eea5becc54eb`, on `pokemon-showdown-mc`. Each old figure is read from the artifact that commit wrote.

Clause	Reg M-B, 7.0.0 (old)	Reg M-C, 1.0.0 (new)	Linkable?
Damage differential	0 of 6000 (<code>bfbf9cf9:data/engine-diff.json</code>)	0 of 6000 (<code>14c36f21:data/engine-diff-regmc.json</code>)	Yes: the same design and seed; each is zero on its own authority
Roster, items	148 of 148 in scope (<code>bfbf9cf9:data/roster.items.json</code>)	166 of 166 (<code>14c36f21:data/roster.items-regmc.json</code>)	No: the in-scope set grew

Clause	Reg M-B, 7.0.0 (old)	Reg M-C, 1.0.0 (new)	Linkable?
			with the regulation
Roster, abilities	196 of 200 in scope (bfbf9cf9:data/roster.abilities.json)	210 of 214 (14c36f21:data/roster.abilities-regmc.json)	No: a different in-scope set
Roster, moves	496 of 497 in scope (bfbf9cf9:data/roster.moves.json)	510 of 511 (14c36f21:data/roster.moves-regmc.json)	No: a different in-scope set
Whole game, lattice A	961 games, 0 parted (bfbf9cf9:data/game-differential.json , --games 1200)	955 games, 0 parted (14c36f21:data/game-differential-regmc.json)	No: a different pool, so different teams
Whole game, lattice B	1069 games, 0 parted (bfbf9cf9:data/game-differential.g1350.json , --games 1350)	1266 games, 0 parted (14c36f21:data/game-differential.g1600-regmc.json)	No
Whole game, lattice C	1497 games, 0 parted (bfbf9cf9:data/game-differential.g1950.json , --games 1950)	1497 games, 0 parted (14c36f21:data/game-differential.g1900-regmc.json)	No: the equal count is a coincidence of two different lattices
Staged mechanics	4632 games, 0 threw (bfbf9cf9:data/all-mechanics-fire.json)	4867 games, 0 threw (14c36f21:data/all-mechanics-fire-regmc.json)	No: a different census drives the staging
Wide held-out draw	no board parted (the readout below)	not run	—

The Reg M-B held-out draw is written under `data/verification/`, where no gate run can overwrite it, so it is quoted as the readout the 7.0.0 edition quoted (CHANGELOG.md 6.83.0):

```
Reg M-B held-out draw, release 0d7b1d9db6d1, --games 12000, pool digest e398641bda45
state.games                                7182
state.games_board_never_diverged           7182  -> board-material 0
```

Impact. Only the damage differential links, and it reads zero on both. Every other old figure answered a question about a different regulation over a different pool, and the right sentence about it is "that answered a question we no longer ask" — not "it became". No model figure from the Reg M-B line survives into this edition: they were withdrawn at 7.0.0, and the models are rebuilt.

6. Mathematics

Matrix game. For payoff matrix M ($m \times n$), the value is $v^* = \max_{\sigma} \min_{\tau} \sigma^T M \tau$ over mixed strategies $\sigma \in \Delta_m$, $\tau \in \Delta_n$. The exploitability of a pair (σ, τ) is $\max_{\tau'} \sigma^T M \tau'$ subtracted from $\max_{\sigma'} \sigma'^T M \tau$, i.e. how much each side could gain by deviating.

Regret matching plus. Each player keeps cumulative regrets $R_t(a) = \max(0, R_{t-1}(a) + u_t(a) - \bar{u}_t)$ and plays in proportion to them; the average strategy's exploitability after T iterations is bounded by a term of order $\Delta(\sqrt{m} + \sqrt{n})/\sqrt{T}$.

Common random numbers. For two cells a, a' scored with the same dice ω_i , $\text{Var}[f(a, \omega) - f(a', \omega)] = \text{Var} f(a) + \text{Var} f(a') - 2 \text{Cov}(f(a), f(a'))$. The covariance is large when the actions differ in a few decisions and share most of the luck, so the difference is far less noisy than with independent dice.

Belief update. For a world w and an observation o , $\beta'(w) \propto \beta(w) \cdot P_MEDICHAM(o \mid S_w, \text{actions})$, with turn order treated as a near-hard constraint away from speed ties.

Restricted Nash Response. The opponent is forced to play the model h with probability p and plays freely otherwise; $p = 0$ is the equilibrium and $p = 1$ a pure best response to h . HYPNO chooses the largest p whose worst case stays within ε of v^* .

Ladder residual. For series s with the Elo expected score $E_s = 1/(1 + 10^{((R_opp - R_me)/400)})$ and result S_s , the per-arm metric is the mean of $S_s - E_s$, read by SPRT; it is insensitive to where on the ladder each series was played.

Sequential probability ratio test. For hypotheses $H_0: \theta = \theta_0$ and $H_1: \theta = \theta_1$ with error rates α and β , stop when the log-likelihood ratio leaves $[\log(\beta/(1 - \alpha)), \log((1 - \beta)/\alpha)]$.

7. Limitations

- **No solver strength figure exists on the certified release.** Everything in §3 is either offline or withheld. The first strength claim will be an arena SPRT on a frozen release.
- **No wide held-out draw on Reg M-C (§4.2).** The lattices are a sample; Reg M-B showed a held-out draw can see what they cannot.
- **Illusion is declared out of scope**, and closed team sheets are out of scope entirely.
- **The mid-turn choice callback is not built.** Until it is, a faint replacement or a pivot's switch-in inside a playout is pre-decided rather than searched.
- **XATU's spread belief is weak (§3.4)**, so speed-order and damage uncertainty stay wide.
- **The prior art is reported, not reproduced.** PokaiTrainer's figures were read through a summary of the paper and are not used as targets.
- **Self-play throughput is unknown** until MEW measures it; the estimates in the research reports span more than an order of magnitude.
- **The ladder is noisy.** Small gains cannot be resolved there in any reasonable time and must be settled offline.

8. References

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- Semantic Versioning 2.0.0; Keep a Changelog 1.1.0.
- Project sources: `solver/PLAN.md` , `solver/LOG.md` , `CHANGELOG-REGMC.md` , `docs/RUNNING-NOTES.md` , `docs/REGMC.md` , and the reports under `docs/_reports/` named in each section.

The plain-English version of this paper is [the deck](#). The commands are in [the technical documentation](#).